Year 9

REVISION PACK

Unit 1

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Unit 1 · Powers of 10 and calculating (Lessons 1.1, 1.2)

Q	Working	Answer
powers of 10		
1.1	**(a)** $0.37 \times 10\,000 = 3700$ **(b)** $5600 \div 1000 = 5.6$	**(a) 3700** **(b) 5.6**
1.2	**(a)** $2.08 \times 1000 = 2080$ **(b)** $49\,000 \div 100\,000 = 0.49$	**(a) 2080** **(b) 0.49**
1.3	**(a)** $0.065 \times 100\,000 = 6500$ **(b)** $730 \div 10\,000 = 0.073$	**(a) 6500** **(b) 0.073**
1.4	**(a)** $1.4 \times 100 = 140$ **(b)** $26\,000 \div 1000 = 26$	**(a) 140** **(b) 26**
1.5	**(a)** $0.009 \times 10\,000 = 90$ **(b)** $85 \div 100 = 0.85$	**(a) 90** **(b) 0.85**
1.6	**(a)** $3.71 \times 100\,000 = 371\,000$ **(b)** $610\,000 \div 1\,000\,000 = 0.61$	**(a) 371 000** **(b) 0.61**
estimating		
1.7	$\approx \dfrac{30 \times 4}{0.5}$ $= \dfrac{120}{0.5}$ $= 240$	**240**
1.8	$\approx \dfrac{80 \times 6}{0.2}$ $= \dfrac{480}{0.2}$ $= 2400$	**2400**

1.9	$\approx \dfrac{10 \times 400}{0.2}$ $= \dfrac{4000}{0.2}$ $= 20\,000$	**20\,000**
1.10	$\approx \dfrac{20 \times 7}{0.4}$ $= \dfrac{140}{0.4}$ $= 350$	**350**
1.11	$\approx \dfrac{500 \times 2}{0.5}$ $= \dfrac{1000}{0.5}$ $= 2000$	**2000**
1.12	$\approx \dfrac{40 \times 7}{0.2}$ $= \dfrac{280}{0.2}$ $= 1400$	**1400**
using a calculator		
1.13	**(a)** $\text{top} = 81.45$ $\text{bottom} = 2.4$ $81.45 \div 2.4 = 33.9375$ **(b)** $33.9375 \approx 33.9$	**(a) 33.9375** **(b) 33.9**
1.14	**(a)** $\text{top} = 63.37$ $\text{bottom} = 4.7$ $63.37 \div 4.7 = 13.48297872$ **(b)** $13.48297872 \approx 13.5$	**(a) 13.48297872** **(b) 13.5**
1.15	**(a)** $\text{top} = 142.68$ $\text{bottom} = 3.3$ $142.68 \div 3.3 = 43.23636364$ **(b)** $43.23636364 \approx 43.2$	**(a) 43.23636364** **(b) 43.2**

1.16	**(a)** $\text{top} = 64.96$ $\text{bottom} = 1.6$ $64.96 \div 1.6 = 40.6$ **(b)** $40.6 \approx 40.6$	**(a) 40.6** **(b) 40.6**
1.17	**(a)** $\text{top} = 189.25$ $\text{bottom} = 5.5$ $189.25 \div 5.5 = 34.40909091$ **(b)** $34.40909091 \approx 34.4$	**(a) 34.40909091** **(b) 34.4**
1.18	**(a)** $\text{top} = 59.07$ $\text{bottom} = 2.9$ $59.07 \div 2.9 = 20.36896552$ **(b)** $20.36896552 \approx 20.4$	**(a) 20.36896552** **(b) 20.4**

Unit 1 · Indices (Lesson 1.3)

Q	Working	Answer
index laws		
2.1	**(a)** $7^{2+5} = 7^{7}$ **(b)** $7^{10-4} = 7^{6}$ **(c)** $7^{5\times 3} = 7^{15}$ **(d)** $7^{3+6} = 7^{9}$ $7^{9-4} = 7^{5}$	**(a)** 7^{7} **(b)** 7^{6} **(c)** 7^{15} **(d)** 7^{5}
2.2	**(a)** $2^{6+3} = 2^{9}$ **(b)** $2^{12-7} = 2^{5}$ **(c)** $2^{4\times 4} = 2^{16}$ **(d)** $2^{5+7} = 2^{12}$ $2^{12-3} = 2^{9}$	**(a)** 2^{9} **(b)** 2^{5} **(c)** 2^{16} **(d)** 2^{9}

2.3	**(a)** $8^{4+9} = 8^{13}$ **(b)** $8^{11-2} = 8^{9}$ **(c)** $8^{2\times 7} = 8^{14}$ **(d)** $8^{9+2} = 8^{11}$ $8^{11-6} = 8^{5}$	**(a)** 8^{13} **(b)** 8^{9} **(c)** 8^{14} **(d)** 8^{5}
2.4	**(a)** $5^{7+2} = 5^{9}$ **(b)** $5^{9-4} = 5^{5}$ **(c)** $5^{3\times 5} = 5^{15}$ **(d)** $5^{4+4} = 5^{8}$ $5^{8-5} = 5^{3}$	**(a)** 5^{9} **(b)** 5^{5} **(c)** 5^{15} **(d)** 5^{3}
2.5	**(a)** $6^{3+8} = 6^{11}$ **(b)** $6^{15-6} = 6^{9}$ **(c)** $6^{7\times 2} = 6^{14}$ **(d)** $6^{8+3} = 6^{11}$ $6^{11-9} = 6^{2}$	**(a)** 6^{11} **(b)** 6^{9} **(c)** 6^{14} **(d)** 6^{2}
2.6	**(a)** $3^{5+6} = 3^{11}$ **(b)** $3^{13-8} = 3^{5}$ **(c)** $3^{4\times 3} = 3^{12}$ **(d)** $3^{2+9} = 3^{11}$ $3^{11-7} = 3^{4}$	**(a)** 3^{11} **(b)** 3^{5} **(c)** 3^{12} **(d)** 3^{4}
zero and negative powers		
2.7	**(a)** $6^{0} = 1$ **(b)** $3^{-2} = \frac{1}{3^{2}}$ $= \frac{1}{9}$	**(a)** **1** **(b)** $\frac{1}{9}$
2.8	**(a)** $25^{0} = 1$ **(b)** $2^{-3} = \frac{1}{2^{3}}$ $= \frac{1}{8}$	**(a)** **1** **(b)** $\frac{1}{8}$

2.9	**(a)** $100^0 = 1$ **(b)** $10^{-2} = \frac{1}{10^2}$ $= \frac{1}{100}$	**(a)** 1 **(b)** $\frac{1}{100}$
2.10	**(a)** $7^0 = 1$ **(b)** $3^{-3} = \frac{1}{3^3}$ $= \frac{1}{27}$	**(a)** 1 **(b)** $\frac{1}{27}$
2.11	**(a)** $15^0 = 1$ **(b)** $6^{-2} = \frac{1}{6^2}$ $= \frac{1}{36}$	**(a)** 1 **(b)** $\frac{1}{36}$
2.12	**(a)** $8^0 = 1$ **(b)** $2^{-4} = \frac{1}{2^4}$ $= \frac{1}{16}$	**(a)** 1 **(b)** $\frac{1}{16}$
a single power and a fraction		
2.13	**(a)** $4 + (-7) = -3$ **(b)** $2^{-3} = \frac{1}{2^3}$ $= \frac{1}{8}$	**(a)** 2^{-3} **(b)** $\frac{1}{8}$
2.14	**(a)** $1 + (-3) = -2$ **(b)** $5^{-2} = \frac{1}{5^2}$ $= \frac{1}{25}$	**(a)** 5^{-2} **(b)** $\frac{1}{25}$
2.15	**(a)** $2 + (-5) = -3$ **(b)** $4^{-3} = \frac{1}{4^3}$ $= \frac{1}{64}$	**(a)** 4^{-3} **(b)** $\frac{1}{64}$

2.16	**(a)** $3 + (-5) = -2$ **(b)** $10^{-2} = \frac{1}{10^2}$ $= \frac{1}{100}$	**(a)** 10^{-2} **(b)** $\frac{1}{100}$
2.17	**(a)** $1 + (-4) = -3$ **(b)** $3^{-3} = \frac{1}{3^3}$ $= \frac{1}{27}$	**(a)** 3^{-3} **(b)** $\frac{1}{27}$
2.18	**(a)** $2 + (-6) = -4$ **(b)** $2^{-4} = \frac{1}{2^4}$ $= \frac{1}{16}$	**(a)** 2^{-4} **(b)** $\frac{1}{16}$

Unit 1 · Standard form (Lessons 1.4, 1.5)

Q	Working	Answer
writing numbers in standard form		
3.1	**(a)** $2.7 \times 10\,000 = 27\,000$ **(b)** 6.3 6.3×10^{-4} **(c)** 8.1 8.1×10^{6}	**(a) 27 000** **(b)** 6.3×10^{-4} **(c)** 8.1×10^{6}
3.2	**(a)** $5.04 \times 1\,000\,000 = 5\,040\,000$ **(b)** 3.9 3.9×10^{-2} **(c)** 4.1 4.1×10^{5}	**(a) 5 040 000** **(b)** 3.9×10^{-2} **(c)** 4.1×10^{5}

3.3	**(a)** $9.1 \times 1000 = 9100$ **(b)** 8.05 8.05×10^{-4} **(c)** 7.2 7.2×10^{7}	**(a) 9100** **(b)** 8.05×10^{-4} **(c)** 7.2×10^{7}
3.4	**(a)** $6.25 \times 1000 = 6250$ **(b)** 5.1 5.1×10^{-3} **(c)** 9.4 9.4×10^{5}	**(a) 6250** **(b)** 5.1×10^{-3} **(c)** 9.4×10^{5}
3.5	**(a)** $1.8 \times 100\,000 = 180\,000$ **(b)** 2.7 2.7×10^{-4} **(c)** 3.6 3.6×10^{6}	**(a) 180 000** **(b)** 2.7×10^{-4} **(c)** 3.6×10^{6}
3.6	**(a)** $7.3 \times 1\,000\,000 = 7\,300\,000$ **(b)** 6.8 6.8×10^{-2} **(c)** 2.5 2.5×10^{7}	**(a) 7 300 000** **(b)** 6.8×10^{-2} **(c)** 2.5×10^{7}
multiplying in standard form		
3.7	$3 \times 7 = 21$ $10^{4+2} = 10^{6}$ 2.1×10^{7}	2.1×10^{7}
3.8	$2.5 \times 3 = 7.5$ $10^{6+5} = 10^{11}$	7.5×10^{11}
3.9	$8 \times 5 = 4E + 1$ $10^{3+7} = 10^{10}$ 4×10^{11}	4×10^{11}

3.10	$1.5 \times 4 = 6$ $10^{9+2} = 10^{11}$	6×10^{11}
3.11	$6 \times 9 = 54$ $10^{2+8} = 10^{10}$ 5.4×10^{11}	5.4×10^{11}
3.12	$2 \times 3.5 = 7$ $10^{5+4} = 10^{9}$	7×10^{9}
dividing in standard form		
3.13	$6.3 \div 3 = 2.1$ $10^{8-2} = 10^{6}$	2.1×10^{6}
3.14	$7.5 \div 2.5 = 3$ $10^{9-4} = 10^{5}$	3×10^{5}
3.15	$4.8 \div 1.6 = 3$ $10^{6-2} = 10^{4}$	3×10^{4}
3.16	$9 \div 4.5 = 2$ $10^{7-3} = 10^{4}$	2×10^{4}
3.17	$1.2 \div 4 = 0.3$ $10^{5-9} = 10^{-4}$ 3×10^{-5}	3×10^{-5}
3.18	$2.4 \div 8 = 0.3$ $10^{10-3} = 10^{7}$ 3×10^{6}	3×10^{6}
ordering		
3.19	smallest power of 10 first	$7.1 \times 10^{-4},\ 4.15 \times 10^{-3},$ $4.2 \times 10^{-3},\ 3.9 \times 10^{-2}$
3.20	smallest power of 10 first	$9.9 \times 10^{4},\ 5.8 \times 10^{5},\ 6 \times 10^{5},$ 1.2×10^{6}
3.21	smallest power of 10 first	$1.1 \times 10^{-3},\ 9.6 \times 10^{-2},$ $2.45 \times 10^{-1},\ 2.5 \times 10^{-1}$
3.22	smallest power of 10 first	$9.2 \times 10^{-4},\ 3.03 \times 10^{-3},$ $3.3 \times 10^{-3},\ 1.7 \times 10^{-2}$

3.23	smallest power of 10 first	$8.1 \times 10^3,\ 5.05 \times 10^4,\ 5.5 \times 10^4,\ 2.2 \times 10^5$
3.24	smallest power of 10 first	$3.9 \times 10^{-6},\ 7.04 \times 10^{-5},\ 7.4 \times 10^{-5},\ 1.6 \times 10^{-4}$
how many times heavier		
3.25	$(1.9 \times 10^{27}) \div (6.42 \times 10^{23})$ $= 2959.502\ldots$	**2960**
3.26	$(1.5 \times 10^5) \div (2 \times 10^{-2})$ $= 7\,500\,000$	**7 500 000**
3.27	$(4.87 \times 10^{24}) \div (3.3 \times 10^{23})$ $= 14.75758\ldots$	**14.8**
3.28	$(5.68 \times 10^{26}) \div (6.42 \times 10^{23})$ $= 884.7352\ldots$	**885**
3.29	$(6 \times 10^3) \div (3 \times 10^{-6})$ $= 2\,000\,000\,000$	**2 000 000 000**
3.30	$(1.02 \times 10^{26}) \div (3.3 \times 10^{23})$ $= 309.0909\ldots$	**309**
how many fit across		
3.31	$2\text{ cm} = 0.02\text{ m}$ $0.02 \div (8 \times 10^{-6}) = 2500$ 2.5×10^3	2.5×10^3
3.32	$4\text{ cm} = 0.04\text{ m}$ $0.04 \div (5 \times 10^{-4}) = 80$ 8×10^1	8×10^1
3.33	$3\text{ cm} = 0.03\text{ m}$ $0.03 \div (6 \times 10^{-5}) = 500$ 5×10^2	5×10^2
3.34	$6\text{ mm} = 0.006\text{ m}$ $0.006 \div (1.5 \times 10^{-10}) = 40\,000\,000$ 4×10^7	4×10^7

3.35	$1.4\ \text{cm} = 0.014\ \text{m}$ $0.014 \div (7 \times 10^{-6}) = 2000$ 2×10^{3}	2×10^{3}
3.36	$5\ \text{cm} = 0.05\ \text{m}$ $0.05 \div (2 \times 10^{-8}) = 2\,500\,000$ 2.5×10^{6}	2.5×10^{6}